

Roc Auc curve

Receiver Operating Characteristic

Characteristic

classification based

models

True Positive Rate (Recall)

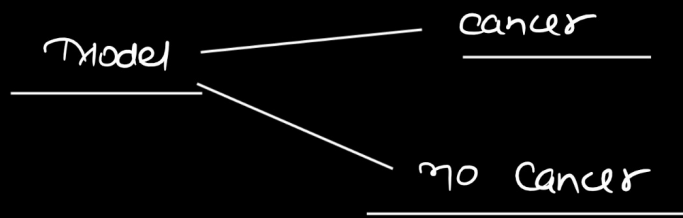
$$\frac{TP}{TP + FN} = TPR$$

Rate (Recall)

fpr $\rightarrow \frac{FP}{FP + TN}$ (false Positive Rate)

	Actual	
	Positive	Negative
Prediction Positive	TP ✓	<u>FP</u>
Prediction Negative	<u>FN</u>	TN ✓

cancer detection model



Auc $\rightarrow$ Area under

the curve

fp $\rightarrow$

Predict = 1
actual = 0 } Predicting cancer when there is none

tells how well the model separates the

& -ve class

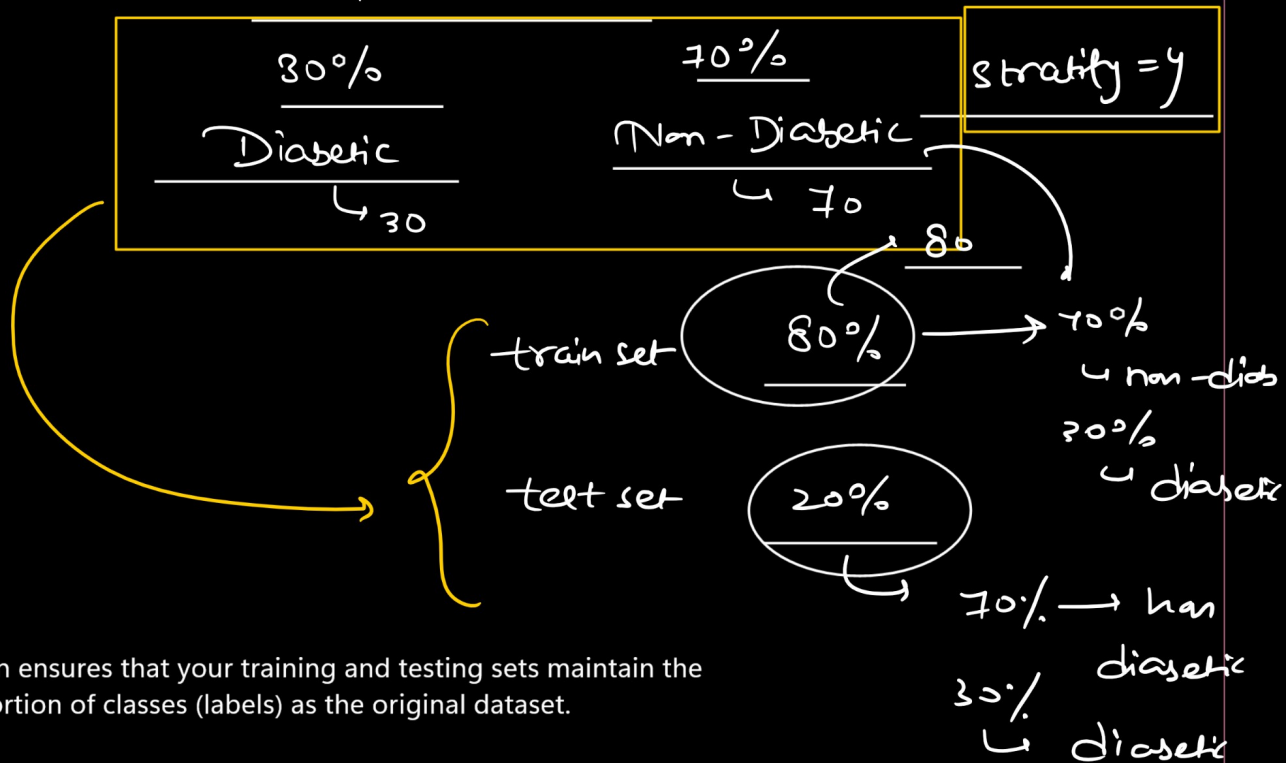
FN $\rightarrow$

Predict = 0
actual = 1 } Missing actual cancer?

Pred

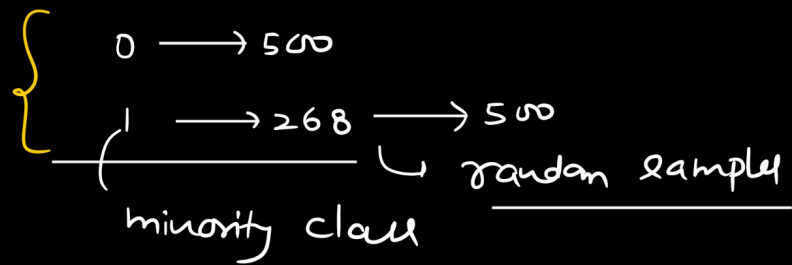
		$\frac{N}{51(TN)}$	$\frac{P}{3(FP)}$
tn, fp, fn, tp	Actual	$\frac{N}{51}$	$\frac{P}{3}$
		$2(FN)$	$87(TP)$

Confusion matrix



stratification ensures that your training and testing sets maintain the same proportion of classes (labels) as the original dataset.

Oversampling



Undersampling

0 → 500 (randomly drop records to make the overall count as 268)

1 → 268